

Selecting Ratio Representations for Comparisons

Choosing a representation: unit-rate comparisons, scaling to a common amount, and tape-diagram totals

Directions:

- You know three ratio tools: **unit rates**, **ratio tables** / **double number lines** (scaling to a common amount), and **tape diagrams** (splitting totals). For each problem, pick the tool that fits and show it in your work space.
- After the warm-up, the problem types are mixed on purpose — part of the job is choosing the method.

- 1. Warm-up.** A box of 6 pencils costs 90 cents. How many cents does one pencil cost?

Answer: _____

- 2. Warm-up.** A cocoa recipe uses 2 cups of cocoa for every 5 cups of milk. Scale the recipe up: how many cups of cocoa go with 20 cups of milk?

Answer: _____

3. Warm-up. Two friends split 42 trading cards in the ratio 3:4. How many cards are in the **larger** share?

Answer: _____

4. Brand A sells 4 pounds of rice for 12 dollars. Brand B sells 5 pounds for 10 dollars.

- (a) Brand A, dollars per pound: _____
- (b) Brand B, dollars per pound: _____
- (c) The cheaper brand per pound is _____

5. Mia reads 20 pages in 8 minutes. Noah reads 27 pages in 12 minutes. Scale both readers to the same 24 minutes to compare them.

- (a) pages Mia reads in 24 minutes: _____
- (b) pages Noah reads in 24 minutes: _____
- (c) the faster reader is _____

6. Team A splits a 60-dollar prize in the ratio 2:3. Team B splits a 63-dollar prize in the ratio 4:5.

- (a) larger share on team A, in dollars: _____
- (b) larger share on team B, in dollars: _____
- (c) Which of those two larger shares is bigger? _____

7. Hiker A covers 9 kilometers every 2 hours. Hiker B covers 12 kilometers every 3 hours. Both hike for 6 hours.

- (a) hiker A's distance in 6 hours, in km: _____
- (b) hiker B's distance in 6 hours, in km: _____
- (c) Who is ahead after 6 hours? _____

8. A pack of 8 yogurts costs 6 dollars (600 cents). A single yogurt costs 80 cents.

- (a) cents per yogurt when buying the pack: _____
- (b) the better buy is _____

9. Shelter A houses cats and dogs in the ratio 5:3, with 16 animals in all. Shelter B houses 9 cats.

(a) number of cats at shelter A: _____

(b) Which shelter has more cats? _____

10. A punch mixes syrup and soda in the ratio 3:5, and a full bowl holds 32 cups of punch. How many cups of syrup are in the bowl? In the work space, solve it with the representation of your choice, and say which representation you chose and why it fits this problem.

Answer: _____

11. The express train covers 210 kilometers in 3 hours. The local train covers 140 kilometers in 2 hours. A classmate says the express train must be faster because 210 is bigger than 140.

- (a) express train speed, km per hour: _____
- (b) local train speed, km per hour: _____
- (c) Write $<$, $>$, or $=$ to compare the speeds: _____

12. Synthesis challenge. Recipe A mixes 2 cups of lemon juice for every 3 cups of water. Recipe B mixes 3 cups of lemon juice for every 5 cups of water.

- (a) Which recipe tastes stronger? (Compare the juice-per-water fractions.) _____
- (b) You make 40 cups (juice plus water) of the **stronger** recipe. How many cups of lemon juice do you need? _____
- (c) In the work space, explain which representation you used for each part.